Area formula for all 3 kinds of triangle

There are 3 common types of triangles (based on sides):

1. Equilateral triangle
2. Isosceles triangle
3. Scalene triangle

Let's see the area formula for each clearly and simply.

1️⃣ Equilateral Triangle

(All three sides equal = a)

- 🔹 Area Formula:

$$\text{Area} = \frac{\sqrt{3}}{4}a^2$$

Where:

- a = side length

✅ Special formula because all sides are equal.

2️⃣ Isosceles Triangle

(Two sides equal)

If equal sides = a

Base = b

- 🔹 Area Formula:

$$\text{Area} = \frac{1}{2} \times b \times h$$

But height h is:

$$h = \sqrt{a^2 - \frac{b^2}{4}}$$

So full formula becomes:

$$\text{Area} = \frac{b}{4}\sqrt{4a^2 - b^2}$$

3️⃣ Scalene Triangle

(All sides different: a, b, c)

We use Heron's Formula

Step 1: Find semi-perimeter

$$s = \frac{a+b+c}{2}$$

Step 2: Area formula

$$\text{Area} = \sqrt{s(s-a)(s-b)(s-c)}$$

🔥 Important: Universal Formula (Works for ALL triangles)

If you know base and height:

$$\text{Area} = \frac{1}{2} \times \text{base} \times \text{height}$$

This works for every triangle if height is known.

If you want, I can also explain area formulas for:

- Right triangle
- Using sine rule
- Using coordinates
- Using vectors

Just tell me 👍

Derivative or proof for all of these

Good question 🔥 — you want the derivation (proof) of area formulas for:

1. Equilateral
2. Isosceles
3. Scalene (Heron's formula)

Let's derive them step-by-step in a clean mathematical way.

🔷 0️⃣ Base Formula (Main Idea)

Every triangle area comes from:

$$\text{Area} = \frac{1}{2} \times \text{base} \times \text{height}$$

All other formulas are derived from this.

1️⃣ Derivation of Equilateral Triangle Area

Let each side = a

Draw a height from the top vertex.

It splits the triangle into two right triangles.

Now:

- Hypotenuse = a
- Base = $a/2$
- Height = h

Using Pythagoras:

$$h^2 + \left(\frac{a}{2}\right)^2 = a^2$$

$$h^2 + \frac{a^2}{4} = a^2$$

$$h^2 = a^2 - \frac{a^2}{4}$$

$$h^2 = \frac{3a^2}{4}$$

$$h = \frac{\sqrt{3}}{2}a$$

Now plug into area formula:

$$\text{Area} = \frac{1}{2} \times a \times \frac{\sqrt{3}}{2}a$$

$$\text{Area} = \frac{\sqrt{3}}{4}a^2$$

✅ Proven.

2️⃣ Derivation of Isosceles Triangle Area

Let equal sides = a

Base = b

Draw height from top vertex.

It splits base into two parts: $b/2$

Now in right triangle:

$$h^2 + \left(\frac{b}{2}\right)^2 = a^2$$

$$h^2 = a^2 - \frac{b^2}{4}$$

$$h = \sqrt{a^2 - \frac{b^2}{4}}$$

Now area:

$$\text{Area} = \frac{1}{2} \times b \times h$$

$$\text{Area} = \frac{1}{2} b \sqrt{a^2 - \frac{b^2}{4}}$$

Multiply inside square root:

$$= \frac{b}{4} \sqrt{4a^2 - b^2}$$

✅ Derived.

3️⃣ Derivation of Scalene Triangle (Heron's Formula)

This one is deeper.

Let sides = a, b, c

Step I: Use cosine rule

$$c^2 = a^2 + b^2 - 2ab\cos C$$

From this:

$$\cos C = \frac{a^2 + b^2 - c^2}{2ab}$$

Now use identity:

$$\sin^2 C = 1 - \cos^2 C$$

After substitution and simplification (long algebra), we get:

$$\sin C = \frac{\sqrt{(a+b+c)(-a+b+c)(a-b+c)(a+b-c)}}{2ab}$$

Now use area formula using sine:

$$\text{Area} = \frac{1}{2}ab\sin C$$

Substitute sin C:

$$\text{Area} = \frac{1}{4}\sqrt{(a+b+c)(-a+b+c)(a-b+c)(a+b-c)}$$

Let:

$$s = \frac{a+b+c}{2}$$

Then the expression becomes:

$$\text{Area} = \sqrt{s(s-a)(s-b)(s-c)}$$

✅ Heron's Formula Proven.

🔥 Summary

All triangle area formulas come from:

$$\text{Area} = \frac{1}{2}bh$$

or

$$\text{Area} = \frac{1}{2}ab\sin C$$

Everything else is algebra manipulation.

If you want, I can also prove:

- Area using vectors
- Area using determinant
- Why sine formula works geometrically
- Full detailed Heron proof without skipping algebra

Tell me how deep you want to go 😎